The Entropic Cost of Counterfactual Consistency: Minimum-Entropy Couplings and Response Trees in Controlled Processes

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Abstract

For a finite controlled stochastic process, a family of intervention policies induces a family of transcript distributions. There are at least three distinct exact Shannon-information costs associated with these distributions. The first, ρ_n, is the largest transcript entropy under a single policy. The second, introduced here and denoted κ_n, is the minimum entropy of one common random source from which each policy transcript can be generated by its own deterministic decoder. Equivalently, κ_n is the ordinary multi-marginal minimum-entropy coupling cost of the policy-indexed transcript laws. The third, χ_n, requires one common random source to admit a single causal response-tree interpretation across all policies. We prove

$$\rho_n \le \kappa_n \le \chi_n,$$

and characterize χ_n as minimum entropy over the subset of transcript couplings supported on tree-compatible tuples. Thus the difference $\Gamma_n := \chi_n - \kappa_n$ measures the extra entropy imposed by counterfactual causal consistency, not merely by sharing one source across marginals.

The constraint is nonredundant. At horizon one, $\Gamma_1 = 0$. At horizon two we give an exact binary example with two adaptive policies for which

$$\Gamma_2 = \frac{3}{8}\log_2 3 \approx 0.59436\,\text{bits}.$$

We then give a second two-stage binary world in which all four deterministic second-stage policies are admissible and prove

$$\Gamma_2 \ge \frac{2}{9}\,\text{bits}.$$

The mechanism is explicit: an unconstrained minimum-entropy coupling may let the same source atom represent different earlier observations under different policy decoders, whereas a response tree must assign a single response to every shared history–action node. The result separates ordinary minimum-entropy coupling from minimum-entropy causal realization in an adaptive controlled interface. We make no claim that constrained couplings, causal transport, or minimum-entropy functional representations are new in themselves; the narrow contribution is the response-tree compatibility formulation and an explicit strict Shannon-entropy separation for policy-indexed controlled processes.

1 Introduction

A controlled stochastic process does not determine one probability distribution but a family of distributions indexed by interventions. If π is a deterministic adaptive policy and T_n^π is the observable transcript through horizon n, write

$$\mu_W^{\pi,n} = \mathcal{L}(T_n^\pi).$$

A previous resource formulation associated two exact costs with this interface [1]. The on-path cost

$$\rho_n(W) = \sup_\pi H(T_n^\pi)$$

charges only the randomness required to generate the transcript under one selected policy. The stronger counterfactual seed cost $\chi_n(W)$ charges a single policy-independent source whose value fixes one deterministic response tree and thereby fixes responses under every admissible intervention.

There is, however, an intermediate construction that the two-cost formulation does not isolate. Suppose one common random variable E is sampled once, but for each policy π we are allowed to use an unrelated deterministic decoder d_π. We require only

$$d_\pi(E) \sim \mu_W^{\pi,n} \qquad \text{for every } \pi.$$

No single causal mechanism is required to explain how the different decoders fit together. Minimizing $H(E)$ is exactly the multi-marginal minimum-entropy coupling problem for the family $\{\mu_W^{\pi,n}\}_\pi$. We denote its value by $\kappa_n(W)$.

This yields a three-level hierarchy:

$$\boxed{\rho_n(W) \;\le\; \kappa_n(W) \;\le\; \chi_n(W).} \tag{1}$$

The first inequality is ordinary deterministic data processing: one common source must be at least as informative as each of its decoded marginals. The second is more structural. A response tree provides a common source for all policies, but not every arbitrary coupling of policy transcripts can be read as one response tree.

The main question is whether this extra causal requirement can have a strict Shannon cost:

$$\Gamma_n(W) := \chi_n(W) - \kappa_n(W) \overset{?}{>} 0. \tag{2}$$

At one time step it cannot: a response tree is simply a list of responses to possible first actions, so an arbitrary coupling of those responses is already a response-tree coupling. Beginning at two time steps, adaptive policies share prefixes and can diverge after observing them. An arbitrary coupling may "relabel" a shared prefix differently under different policy coordinates; a single response tree cannot.

We establish four elementary results.

(i) We prove that κ_n is exactly the ordinary multi-marginal minimum-entropy coupling of the policy-indexed transcript laws.

(ii) We characterize χ_n as the same entropy minimization restricted to couplings supported on *tree-compatible* transcript tuples.

(iii) We prove $\Gamma_1 = 0$ and give a two-policy, two-stage binary example with the exact strict gap $\Gamma_2 = (3/8)\log_2 3$ bits.

(iv) We show that the phenomenon is not an artifact of restricting the policy family: a second binary example with all four deterministic second-stage policies satisfies $\Gamma_2 \ge 2/9$ bit.

Minimum-entropy couplings are classical optimization objects [2, 3]; they have also been used constructively to encode information into Markov-decision-process trajectories [4]; their connection with minimum-entropy exogenous variables and functional representations is central in entropic causal inference [5, 6]. Sequential innovation representations also study stochastic processes driven by simpler exogenous innovations [7]. Separately, causal optimal transport restricts transport plans by nonanticipativity or filtration constraints [8], and contextuality frameworks study when context-indexed random variables admit joint couplings satisfying additional cross-context requirements [9]. Accordingly, no novelty is claimed for the broad idea of imposing compatibility constraints on couplings. The specific object here is narrower: Shannon-entropy minimization over the coupling polytope of *policy transcripts*, with compatibility defined by a single deterministic response tree. The strict examples show that this restriction is not merely a rephrasing of ordinary minimum-entropy coupling.

2 Finite controlled interfaces

Fix finite action and observation alphabets $\mathcal{A}$ and $\mathcal{O}$. At time t, a history is

$$h_{t-1} = (a_1, o_1, \ldots, a_{t-1}, o_{t-1}), \qquad h_0 = \varnothing.$$

A deterministic policy π specifies an action

$$a_t = \pi_t(h_{t-1}).$$

A finite-horizon controlled world W is specified by causal kernels

$$P_W(o_t \mid h_{t-1}, a_t), \qquad 1 \le t \le n. \tag{3}$$

For policy π, the induced transcript is

$$T_n^\pi = (A_1, O_1, \ldots, A_n, O_n)$$

with law $\mu_W^{\pi,n}$ on the finite transcript space $\mathcal{T}_n$.

Let $\mathcal{G} \subseteq \Pi_n$ be a nonempty finite family of admissible deterministic policies. Working with a general $\mathcal{G}$ is useful because an experimental interface may expose only a specified intervention family. When $\mathcal{G} = \Pi_n$, the superscript $\mathcal{G}$ will be omitted.

Definition 1 (On-path cost)**.** *For an admissible policy family $\mathcal{G}$, define*

$$\rho_n^{\mathcal{G}}(W) := \max_{\pi \in \mathcal{G}} H(T_n^\pi). \tag{4}$$

All entropies are Shannon entropies in bits.

2.1 Response trees

A deterministic depth-n response tree assigns one observation to every history–action node that can occur through depth n. Let $\mathcal{R}_n$ denote the finite set of such trees. Given a tree r and a policy π, recursion through the tree produces a unique transcript, denoted

$$F_{\pi,n}(r) \in \mathcal{T}_n.$$

For an admissible family $\mathcal{G}$, let

$$\mathcal{Q}_n^{\mathcal{G}}(W) := \left\{ q \in \Delta(\mathcal{R}_n) : (F_{\pi,n})_\# q = \mu_W^{\pi,n} \quad \forall \pi \in \mathcal{G} \right\}. \tag{5}$$

The set is nonempty for the kernel model in Eq. (3); for example, one may independently sample a response at every history–action node according to the corresponding kernel.

Definition 2 (Counterfactual response-tree cost)**.** *Define*

$$\chi_n^{\mathcal{G}}(W) := \min_{q \in \mathcal{Q}_n^{\mathcal{G}}(W)} H_q(R_n), \tag{6}$$

where $R_n \sim q$.

Operationally, $\chi_n^{\mathcal{G}}$ is the minimum entropy of one policy-independent exogenous variable that, once fixed, determines a causal response mechanism reproducing all policy laws in $\mathcal{G}$ [1].

3 The policywise common-source cost

We now insert an intermediate notion between ρ_n and χ_n.

Definition 3 (Policywise common-source cost)**.** *Let $\kappa_n^{\mathcal{G}}(W)$ be the minimum $H(E)$ over all finite random variables E for which, for each $\pi \in \mathcal{G}$, there exists a deterministic map d_π satisfying*

$$d_\pi(E) \sim \mu_W^{\pi,n}. \tag{7}$$

The maps d_π are not required to arise from one causal mechanism.

The distinction is important. The same source atom e may be decoded to one earlier observation under policy π and to a different earlier observation under policy σ. This is harmless for marginal generation but may be impossible under a single causal response tree.

3.1 Equivalence with ordinary minimum-entropy coupling

For a finite family of distributions $\{\mu_\pi\}_{\pi\in\mathcal{G}}$, write

$$\mathcal{C}(\mu_\pi : \pi \in \mathcal{G}) := \big\{P \in \Delta(\mathcal{T}_n^{\mathcal{G}}) : P_\pi = \mu_\pi \quad \forall \pi \in \mathcal{G}\big\} \tag{8}$$

for the ordinary multi-marginal coupling polytope.

Proposition 1 (Common source = minimum-entropy coupling)**.** *For every finite W, n, and $\mathcal{G}$,*

$$\boxed{\kappa_n^{\mathcal{G}}(W) = \min_{P\in\mathcal{C}(\mu_W^{\pi;n}:\pi\in\mathcal{G})} H_P\big((X_\pi)_{\pi\in\mathcal{G}}\big).} \tag{9}$$

Proof. Suppose E and decoders d_π satisfy Eq. (7). Define $X_\pi = d_\pi(E)$. Then $(X_\pi)_{\pi\in\mathcal{G}}$ is a coupling of the required marginals and is a deterministic function of E. Hence

$$H((X_\pi)_\pi) \le H(E).$$

Taking the infimum over common sources gives the "$\le$" direction in Eq. (9).

Conversely, for any coupling $(X_\pi)_{\pi\in\mathcal{G}}$ with the required marginals, take

$$E = (X_\pi)_{\pi\in\mathcal{G}}$$

and let d_π be the coordinate projection. Then $H(E) = H((X_\pi)_\pi)$ and every marginal is reproduced. Minimizing proves equality. $\square$

Thus κ_n is not a new coupling primitive. It is exactly the ordinary minimum-entropy coupling problem [2, 3] applied to the family of policy transcript laws.

Theorem 1 (Three-cost hierarchy)**.** *For every finite controlled world, horizon, and admissible policy family,*

$$\boxed{\rho_n^{\mathcal{G}}(W) \le \kappa_n^{\mathcal{G}}(W) \le \chi_n^{\mathcal{G}}(W).} \tag{10}$$

Proof. If $d_\pi(E) \sim T_n^\pi$, then deterministic data processing gives $H(T_n^\pi) \le H(E)$ for every π. Maximizing over π and minimizing over E gives $\rho_n^{\mathcal{G}} \le \kappa_n^{\mathcal{G}}$.

For the second inequality, let $R_n \sim q$ be any feasible response-tree seed. Then

$$X_\pi := F_{\pi,n}(R_n)$$

defines a coupling of all policy transcript laws. Hence, by Proposition 1,

$$\kappa_n^{\mathcal{G}} \le H((X_\pi)_\pi) \le H(R_n).$$

Minimizing over $q \in \mathcal{Q}_n^{\mathcal{G}}(W)$ yields $\kappa_n^{\mathcal{G}} \le \chi_n^{\mathcal{G}}$. $\square$

Proposition 2 (Monotonicity under policy refinement)**.** *If $\mathcal{G}_1 \subseteq \mathcal{G}_2$, then*

$$\rho_n^{\mathcal{G}_1} \le \rho_n^{\mathcal{G}_2}, \qquad \kappa_n^{\mathcal{G}_1} \le \kappa_n^{\mathcal{G}_2}, \qquad \chi_n^{\mathcal{G}_1} \le \chi_n^{\mathcal{G}_2}.$$

Proof. The first statement follows from maximizing over a larger set. The second follows because a coupling for more marginals projects to a coupling for fewer marginals. The third follows because adding policies adds constraints to Eq. (5). $\square$

4 Response-tree compatibility as a support constraint

The previous hierarchy can be sharpened into an exact constrained-coupling statement.

Definition 4 (Tree-compatible transcript tuple)**.** *A tuple*

$$x = (t^\pi)_{\pi\in\mathcal{G}} \in \mathcal{T}_n^{\mathcal{G}}$$

is tree-compatible *if there exists a deterministic response tree $r \in \mathcal{R}_n$ such that*

$$F_{\pi,n}(r) = t^\pi \qquad \forall \pi \in \mathcal{G}.$$

Let $\mathsf{TC}_n(\mathcal{G}) \subseteq \mathcal{T}_n^{\mathcal{G}}$ denote the set of all such tuples.

The definition has a simple local characterization.

Lemma 1 (Common-node consistency)**.** *A tuple $(t^\pi)_{\pi\in\mathcal{G}}$ of policy-consistent transcripts is tree-compatible if and only if the following holds: whenever two transcript coordinates visit the same history–action node (h_{t-1}, a_t), they record the same next observation there.*

Proof. Necessity is immediate because a deterministic response tree assigns one observation to each history–action node.

For sufficiency, assign to every history–action node visited by at least one coordinate the observation recorded by that coordinate. The hypothesis makes this assignment well-defined when several coordinates visit the same node. Assign arbitrary observations to all remaining nodes. The resulting response tree reproduces every transcript in the tuple. □

Thus tree compatibility is a cross-policy prefix constraint. It does not ask that policy coordinates be independent or identical; it asks only that they cannot disagree about the response of a node that they claim to share.

Define the tree-compatible coupling set

$$\mathcal{C}^{\mathcal{G}}_{\mathrm{tree}}(W) := \left\{P \in \mathcal{C}(\mu^{\pi,n}_W : \pi \in \mathcal{G}) : P\big(\mathsf{TC}_n(\mathcal{G})\big) = 1\right\}. \tag{11}$$

Theorem 2 (Response-tree cost as constrained MEC)**.** *For every finite $W, n,$ and $\mathcal{G}$,*

$$\boxed{\chi^{\mathcal{G}}_n(W) = \min_{P\in\mathcal{C}^{\mathcal{G}}_{\mathrm{tree}}(W)} H_P\big((X_\pi)_{\pi\in\mathcal{G}}\big).} \tag{12}$$

Proof. Let $R_n \sim q \in \mathcal{Q}^{\mathcal{G}}_n(W)$ and define

$$\Phi(R_n) := (F_{\pi,n}(R_n))_{\pi\in\mathcal{G}}.$$

The law of $\Phi(R_n)$ belongs to $\mathcal{C}^{\mathcal{G}}_{\mathrm{tree}}(W)$, and

$$H(\Phi(R_n)) \le H(R_n).$$

Therefore the right-hand side of Eq. (12) is no larger than $\chi^{\mathcal{G}}_n$.

Conversely, let $P \in \mathcal{C}^{\mathcal{G}}_{\mathrm{tree}}(W)$. Because all spaces are finite, choose for each $x \in \mathsf{TC}_n(\mathcal{G})$ one response tree $s(x)$ such that $\Phi(s(x)) = x$. The map s is injective on $\mathsf{TC}_n(\mathcal{G})$: if $s(x) = s(x')$, applying Φ gives $x = x'$. If $X \sim P$ and $R_n = s(X)$, then

$$H(R_n) = H(X),$$

and R_n reproduces every required policy marginal. Thus the induced law of R_n belongs to $\mathcal{Q}^{\mathcal{G}}_n(W)$, so

$$\chi^{\mathcal{G}}_n(W) \le H(X).$$

Minimizing over P proves equality. □

The only difference between κ_n and χ_n is therefore the feasible coupling set:

$$\begin{aligned} \kappa^{\mathcal{G}}_n &= \min_{P\in\mathcal{C}} H(P), \\ \chi^{\mathcal{G}}_n &= \min_{P\in\mathcal{C}:\ \mathrm{supp}\,P\subseteq\mathsf{TC}_n(\mathcal{G})} H(P). \end{aligned} \tag{13}$$

Definition 5 (Causal-consistency entropy premium)**.** *Define*

$$\boxed{\Gamma^{\mathcal{G}}_n(W) := \chi^{\mathcal{G}}_n(W) - \kappa^{\mathcal{G}}_n(W) \ge 0.} \tag{14}$$

The original counterfactual premium $\Delta^{\mathrm{cf}}_n := \chi_n - \rho_n$ therefore decomposes exactly as

$$\boxed{\Delta^{\mathrm{cf}}_n = \underbrace{(\kappa_n - \rho_n)}_{\text{common-source premium}} + \underbrace{(\chi_n - \kappa_n)}_{\text{causal-consistency premium}}.} \tag{15}$$

The first term is the cost of using one source rather than a policy-specific source. The second is the additional cost of requiring that the common source possess one coherent counterfactual causal interpretation.

4.1 Why no premium appears at one step

Proposition 3 (Horizon-one collapse)**.** *For every one-step controlled world and every policy family,*

$$\boxed{\kappa_1^{\mathcal{G}}(W) = \chi_1^{\mathcal{G}}(W),} \qquad \Gamma_1^{\mathcal{G}}(W) = 0. \tag{16}$$

Proof. At one step a deterministic policy chooses a single initial action. A response tree is therefore exactly a tuple specifying one observation for each action that occurs in $\mathcal{G}$. There are no shared nontrivial histories at which policies can diverge after observing a common prefix. Consequently every ordinary coupling of the policy outcomes is tree-compatible, so $\mathcal{C}_{\text{tree}}^{\mathcal{G}} = \mathcal{C}$ and Theorem 2 gives equality. □

Thus any strict causal-consistency premium is intrinsically sequential.

5 A branchwise formula for two-stage worlds

The separation examples use a simple two-stage template. At stage one there is only one action and the world emits Y in a finite alphabet $\mathcal{Y}$. At stage two, after observing $Y = y$, the policy chooses an action a and the world emits a final variable Z with law

$$\nu_{y,a} = \mathcal{L}(Z \mid Y = y, A = a).$$

Let

$$\mathcal{A}_y(\mathcal{G}) := \{\pi_2(y) : \pi \in \mathcal{G}\}$$

be the actions actually exposed by the policy family at branch y. For a finite family of marginal distributions $\{\nu_j\}_{j\in J}$, write

$$\mathsf{MEC}(\nu_j : j \in J) := \min H((V_j)_{j\in J}) \tag{17}$$

over all couplings with $V_j \sim \nu_j$.

Lemma 2 (Branchwise response-tree formula)**.** *For the two-stage template above,*

$$\boxed{\chi_2^{\mathcal{G}}(W) = H(Y) + \sum_{y\in\mathcal{Y}} P(Y = y)\,\mathsf{MEC}(\nu_{y,a} : a \in \mathcal{A}_y(\mathcal{G})).} \tag{18}$$

Proof. A response tree first fixes Y. Conditional on $Y = y$, it must also fix a joint vector of second-stage responses $V_y = (Z_{y,a})_{a\in\mathcal{A}_y(\mathcal{G})}$ whose marginals are the laws $\nu_{y,a}$. Therefore every feasible tree satisfies

$$H(R_2) \geq H(Y) + H(V_Y \mid Y) \geq H(Y) + \sum_y P(Y = y)\mathsf{MEC}(\nu_{y,a} : a \in \mathcal{A}_y(\mathcal{G})).$$

For achievability, conditional on $Y = y$ sample an entropy-minimizing coupling for V_y, and set all responses on branches $y' \neq y$ to fixed canonical values. This defines a feasible response tree with equality throughout. □

For two Bernoulli marginals, the branchwise minimum is elementary. If $U \sim \text{Bern}(p)$ and $V \sim \text{Bern}(q)$, their couplings are parameterized by $t = P(U = 1, V = 1)$ on the Fréchet interval

$$\max\{0, p + q - 1\} \leq t \leq \min\{p, q\}.$$

Shannon entropy is concave in the joint probability vector, so its minimum on this interval occurs at an endpoint.

6 Exact strict separation with two adaptive policies

We now exhibit the smallest conceptual mechanism behind $\Gamma_2 > 0$.

Let

$$P(Y = 0) = P(Y = 1) = \frac{1}{2}. \tag{19}$$

After observing Y, the agent chooses $A \in \{0, 1\}$, and the final output $Z \in \{0, 1\}$ satisfies, for both values of y,

$$\begin{aligned} P(Z = 1 \mid Y = y, A = 0) &= \frac{1}{4}, \\ P(Z = 1 \mid Y = y, A = 1) &= \frac{1}{2}. \end{aligned} \tag{20}$$

Take two deterministic adaptive policies

$$\pi = (0, 1), \qquad \sigma = (1, 0), \tag{21}$$

where (a_0, a_1) means action a_y is selected after observing $Y = y$. Thus π selects $A = Y$ and σ selects $A = 1 - Y$.

Because the action is determined by (Y, π), it suffices to record the reduced transcript (Y, Z). In the outcome order

$$(0, 1), (0, 0), (1, 1), (1, 0),$$

the two transcript laws are

$$\begin{aligned} \mu^\pi &= \frac{1}{8}(1, 3, 2, 2), \\ \mu^\sigma &= \frac{1}{8}(2, 2, 1, 3). \end{aligned} \tag{22}$$

They are permutations of one another.

Theorem 3 (Exact two-policy causal premium)**.** *For the world in Eqs.* (19)–(21) *and policy family* $\mathcal{G} = \{\pi, \sigma\}$,

$$\kappa_2^{\mathcal{G}}(W) = \frac{5}{2} - \frac{3}{8}\log_2 3, \tag{23}$$

$$\chi_2^{\mathcal{G}}(W) = \frac{5}{2}, \tag{24}$$

and therefore

$$\boxed{\Gamma_2^{\mathcal{G}}(W) = \frac{3}{8}\log_2 3 \approx 0.59436094\,\text{bits}.} \tag{25}$$

Proof. First consider $\kappa_2^{\mathcal{G}}$. Both marginals in Eq. (22) have the same entropy, namely

$$H(\mu^\pi) = H(\mu^\sigma) = H\left(\frac{3, 2, 2, 1}{8}\right) = \frac{5}{2} - \frac{3}{8}\log_2 3. \tag{26}$$

Every common source must have entropy at least that of each deterministic decoding, so

$$\kappa_2^{\mathcal{G}} \geq H(\mu^\pi).$$

Conversely, take a source E with atom probabilities

$$P_E = \frac{1}{8}(3, 2, 2, 1).$$

Because μ^π and μ^σ are permutations of this same probability vector, deterministic relabellings of the four source atoms generate each marginal exactly. Hence

$$\kappa_2^{\mathcal{G}} \leq H(E) = H(\mu^\pi),$$

proving Eq. (23).

Now consider $\chi_2^{\mathcal{G}}$. At either value $Y = y$, the two policies expose both actions. A response tree must therefore couple

$$Z_0 \sim \mathrm{Bern}\left(\frac{1}{4}\right), \qquad Z_1 \sim \mathrm{Bern}\left(\frac{1}{2}\right).$$

Writing $t = P(Z_0 = 1, Z_1 = 1)$, the joint atom probabilities are

$$t, \quad \frac{1}{4} - t, \quad \frac{1}{2} - t, \quad \frac{1}{4} + t, \qquad 0 \leq t \leq \frac{1}{4}.$$

By concavity of entropy the minimum occurs at $t = 0$ or $t = 1/4$, and in either case the nonzero atom probabilities are

$$\frac{1}{2}, \frac{1}{4}, \frac{1}{4},$$

with entropy $3/2$ bits. Lemma 2 now gives

$$\chi_2^{\mathcal{G}} = H(Y) + \frac{1}{2}\cdot\frac{3}{2} + \frac{1}{2}\cdot\frac{3}{2} = 1 + \frac{3}{2} = \frac{5}{2}.$$

Subtracting Eq. (23) proves Eq. (25). $\square$

Remark 1 (Where the saved entropy comes from)**.** *An unconstrained common source is allowed to use a source atom e that decodes to $Y = 0$ under d_π but to $Y = 1$ under d_σ. A response tree cannot do this. Both policies take the same unique first action, so by Lemma 1 they must record the same first observation on each joint source atom. The entropy gap in Theorem 3 is therefore a cost of preserving a common causal prefix across counterfactual policy coordinates.*

7 Strict separation with the full deterministic policy family

The preceding example uses two selected adaptive policies. We next show that strict separation persists when *all* deterministic second-stage policies are included.

Let

$$P(Y = 0) = \frac{4}{9}, \qquad P(Y = 1) = \frac{5}{9}, \tag{27}$$

and let the second-stage binary output satisfy

$$\begin{array}{c|cc} & A = 0 & A = 1 \\ \hline Y = 0 & P(Z = 1) = \frac{1}{4} & P(Z = 1) = \frac{1}{2} \\ Y = 1 & P(Z = 1) = \frac{1}{5} & P(Z = 1) = \frac{2}{5}. \end{array} \tag{28}$$

There are four deterministic policies

$$\pi_{00}, \pi_{01}, \pi_{10}, \pi_{11},$$

where π_{ij} chooses action i after $Y = 0$ and action j after $Y = 1$. In the same reduced-transcript outcome order as before, their laws are

$$\begin{aligned} \mu^{00} &= \frac{1}{9}(1, 3, 1, 4), \\ \mu^{01} &= \frac{1}{9}(1, 3, 2, 3), \\ \mu^{10} &= \frac{1}{9}(2, 2, 1, 4), \\ \mu^{11} &= \frac{1}{9}(2, 2, 2, 3). \end{aligned} \tag{29}$$

Theorem 4 (Full-policy strict separation)**.** *For the world in Eqs.* (27)–(28) *with the full policy family* $\Pi_2 = \{\pi_{00}, \pi_{01}, \pi_{10}, \pi_{11}\}$,

$$\chi_2(W) = \frac{5}{3}\log_2 3 - \frac{2}{9}, \tag{30}$$

while

$$\kappa_2(W) \le \frac{5}{3}\log_2 3 - \frac{4}{9}. \tag{31}$$

Consequently,

$$\boxed{\Gamma_2(W) = \chi_2(W) - \kappa_2(W) \ge \frac{2}{9}\text{ bits} > 0.} \tag{32}$$

Proof. We first compute χ_2. At branch $Y = 0$, a response tree must couple Bern(1/4) and Bern(1/2). As in the proof of Theorem 3, the minimum conditional joint entropy has nonzero weights

$$\frac{1}{2}, \frac{1}{4}, \frac{1}{4}.$$

After multiplication by $P(Y = 0) = 4/9$, these contribute unconditional tree-atom weights

$$\frac{2}{9}, \frac{1}{9}, \frac{1}{9}.$$

At branch $Y = 1$, the required marginals are Bern(1/5) and Bern(2/5). The two Fréchet endpoints have nonzero weight multisets

$$\left\{\frac{2}{5}, \frac{2}{5}, \frac{1}{5}\right\} \quad \text{and} \quad \left\{\frac{3}{5}, \frac{1}{5}, \frac{1}{5}\right\}.$$

The second is more concentrated and has smaller entropy. Multiplying by $P(Y = 1) = 5/9$ gives

$$\frac{3}{9}, \frac{1}{9}, \frac{1}{9}.$$

Lemma 2 therefore yields an optimal reduced response-tree probability vector

$$\frac{1}{9}(3,2,1,1,1,1), \tag{33}$$

so

$$\chi_2 = H\left(\frac{3,2,1,1,1,1}{9}\right) = \log_2 9 - \frac{1}{9}\big(3\log_2 3 + 2\log_2 2\big) = \frac{5}{3}\log_2 3 - \frac{2}{9}.$$

It remains to construct an unconstrained common source with strictly smaller entropy. Let E have five atoms $e_1, \ldots, e_5$ with integer weights

$$(3,2,2,1,1)/9. \tag{34}$$

For each policy, partition these source atoms into the four reduced transcript outcomes as shown in Table 1. The entries are source weights; a sum means that both corresponding source atoms decode to the same transcript outcome.

Table 1: Deterministic decoder partitions for the common source $(3,2,2,1,1)/9$. Columns are the reduced outcomes $(0,1),(0,0),(1,1),(1,0)$.

policy	$(0,1)$	$(0,0)$	$(1,1)$	$(1,0)$
π_{00}	1	3	1	$2+2$
π_{01}	1	3	2	$2+1$
π_{10}	2	2	1	$3+1$
π_{11}	2	2	$1+1$	3

For example, one concrete assignment is obtained by taking source-atom weights $w(e_1), \ldots, w(e_5) = (3,2,2,1,1)$ and using

$$\begin{aligned}
\pi_{00}: &\quad e_4 \mid e_1 \mid e_5 \mid (e_2, e_3),\\
\pi_{01}: &\quad e_4 \mid e_1 \mid e_2 \mid (e_3, e_5),\\
\pi_{10}: &\quad e_2 \mid e_3 \mid e_4 \mid (e_1, e_5),\\
\pi_{11}: &\quad e_2 \mid e_3 \mid (e_4, e_5) \mid e_1,
\end{aligned}$$

where vertical bars separate the four transcript outcomes. The induced marginal weights are exactly those in Eq. (29). Hence

$$\kappa_2 \le H(E) = H\left(\frac{3,2,2,1,1}{9}\right) = \log_2 9 - \frac{1}{9}\big(3\log_2 3 + 4\log_2 2\big) = \frac{5}{3}\log_2 3 - \frac{4}{9}.$$

Subtracting this upper bound from the exact value of χ_2 gives Eq. (32). □

The proof does not require knowing the exact value of κ_2 in the full-policy example. An explicit feasible unconstrained coupling whose entropy lies below the exact tree-compatible optimum is sufficient to establish strict separation.

8 What the separation means

Theorem 2 isolates the source of the difference. An ordinary minimum-entropy coupling is free to coordinate complete transcript variables in any way compatible with their marginals. It has no obligation to interpret the coordinates as different interventions applied to one common dynamical object. A response-tree coupling has precisely that obligation.

In particular, if two policies share a history–action node, then their joint coupling cannot assign two different responses to that node on the same source atom. The constraint is counterfactual: only one policy is run in any one experiment, but the source is required to define what would have happened under all policies in a mutually compatible manner.

This distinguishes three operational questions.

(a) *Policy-specific generation:* how much entropy is needed if a different source may be chosen after the policy is fixed? This gives ρ_n.

(b) *Common-source marginal generation:* how much entropy is needed if one source must support every policy marginal, but each policy may use an unrelated decoder? This gives κ_n.

(c) *Common causal realization:* how much entropy is needed if the same source must instantiate one deterministic causal response object from which every policy transcript is read? This gives χ_n.

The strict examples prove that (b) and (c) are inequivalent. Therefore a minimum-entropy coupling of policy marginals is, in general, *not* a minimum-entropy causal realization of the controlled interface.

9 Relation to existing coupling frameworks and novelty boundary

The mathematical ingredients surrounding the result have substantial prior art, and the boundary is worth stating explicitly.

Minimum-entropy coupling. Entropy minimization over couplings with fixed marginals is well established. Kovačević, Stanojević, and Šenk introduced and analyzed minimum entropy coupling and proved hardness results [2]. Cicalese, Gargano, and Vaccaro developed approximation results and multi-marginal extensions [3]. Proposition 1 deliberately places κ_n inside this existing theory rather than presenting it as a new primitive. Minimum-entropy coupling has also been used inside controlled dynamics, for example to communicate exogenous information through Markov decision process trajectories [4]. That problem optimizes message–trajectory couplings; it does not compare arbitrary cross-policy transcript couplings with the subset admitting one response-tree semantics.

Minimum-entropy exogenous variables and functional representations. Entropic causal inference relates minimum-entropy exogenous variables to minimum-entropy couplings [5]. More recent functional representation work gives converse bounds for the same equivalence [6]. Innovation representations of stochastic processes also study how a process can be driven by simpler exogenous innovations [7]. The present distinction is not between observed and exogenous variables in one fixed structural equation; it is between arbitrary joint couplings of *policy transcript laws* and those admitting one joint response-tree semantics across interventions.

Causal transport. Causal and adapted optimal transport restrict couplings by temporal information-flow constraints and establish a substantial theory of such restrictions [8]. Hence the general statement "causal constraints can change a coupling optimization problem" is not new. Our constraint and objective are different: the marginals are complete policy-indexed transcript laws, the cost is their joint Shannon entropy, and feasibility is support on the image of a deterministic counterfactual response tree.

Contextual couplings. Contextuality-by-Default and related coupling approaches study context-indexed random variables and ask whether their observed bunch distributions can be embedded into a joint coupling satisfying prescribed cross-context connection properties [9]. There is a family resemblance: policy labels act as contexts, and response-tree compatibility identifies responses at shared causal nodes. The present construction differs in two ways relevant here. First, a response-tree coupling always exists for the classical kernel model; the issue is not existence versus contextuality but the *minimum entropy* attainable under the causal support constraint. Second, the connection structure is generated recursively by adaptive histories rather than supplied as a static context–content incidence pattern.

Claim boundary. No claim is made that minimum-entropy coupling, constrained coupling, causal transport, functional representation, or contextual coupling is novel. The narrow result established here is:

> For policy-indexed transcript laws of a finite controlled process, the minimum Shannon entropy of an arbitrary common-source coupling can be strictly smaller than the minimum Shannon entropy of a coupling that admits a single counterfactual response-tree realization.

Theorems 3 and 4 provide explicit finite rational witnesses. A literature search cannot establish absolute priority, so the appropriate novelty claim is structural and framework-relative rather than historical.

10 Consequences for exact resource accounting

The separation sharpens the interpretation of the response-tree cost χ_n. If one only observes that the family of policy transcript laws has a low-entropy common coupling, one has not yet shown that a physical policy-independent source of that entropy can drive one causal realization. The coupling may be using policy-dependent reinterpretations of shared history.

In the exact common-seed resource model of [1], a physical seed E is fixed before the policy and subsequent evolution is deterministic and causal. Such a seed determines one response tree, so the correct lower bound remains

$$H(E) \geq \chi_n,$$

not merely

$$H(E) \geq \kappa_n.$$

Theorem 4 shows that replacing χ_n by an unconstrained MEC of the policy marginals can strictly understate the required common-seed entropy.

Conversely, κ_n remains operationally meaningful. It quantifies the least entropy needed when policy identity is available to an arbitrary decoder that need not preserve cross-policy causal semantics. This can model, for example, a library of policy-specific deterministic generators sharing one compressed random source. The distinction is therefore not "physical" versus "mathematical" but two different operational contracts on the decoder.

Equation (15) also refines the earlier counterfactual premium. A positive value of $\chi_n - \rho_n$ can arise from two logically different sources:

(i) pooling all policy marginals into one random source already costs extra entropy, $\kappa_n - \rho_n$;

(ii) imposing one shared causal semantics costs additional entropy, $\Gamma_n = \chi_n - \kappa_n$.

The second term is exactly what ordinary MEC does not see.

11 Open problems

The separation raises several concrete questions.

(1) **Zero-premium characterization.** Give necessary and sufficient conditions on a controlled interface for $\Gamma_n^{\mathcal{G}} = 0$. Horizon one is trivial; the interesting question is which sequential interfaces admit an entropy-minimizing coupling supported on $\mathsf{TC}_n(\mathcal{G})$.

(2) **Exact complexity.** Ordinary MEC is NP-hard [2, 5]. Since $\chi_1 = \kappa_1$, exact response-tree minimization inherits hardness in general encodings. The complexity of deciding whether $\Gamma_n > 0$, or approximating the premium directly, remains to be characterized.

(3) **Asymptotic behavior.** Determine whether causal-consistency premiums can grow linearly with horizon, whether normalized limits exist under stationary assumptions, and under what product constructions the premium is additive or subadditive.

(4) **Approximate realization.** Replace exact policy marginals by total-variation or other fidelity balls and compare unconstrained approximate MEC with approximate tree-compatible MEC. The interaction between causal consistency and approximation may differ sharply from the exact case.

(5) **Algorithms exploiting tree structure.** The feasible set in Eq. (11) is a coupling polytope intersected with a combinatorial support relation generated recursively by histories. Dynamic programming, branch decomposition, or adapted transport methods may exploit this structure more directly than generic multi-marginal MEC algorithms.

12 Conclusion

A family of intervention-indexed transcript distributions admits two distinct notions of a shared random source. An unconstrained source may be decoded separately for each policy; its minimum entropy is the ordinary multi-marginal MEC value κ_n. A causal source must instantiate one response tree whose branches jointly encode all policy counterfactuals; its minimum entropy is χ_n.

The difference is exactly a support restriction on the coupling problem:

$$\kappa_n = \min_{P \in \mathcal{C}} H(P), \qquad \chi_n = \min_{P \in \mathcal{C},\ \mathrm{supp}(P) \subseteq \mathsf{TC}_n} H(P).$$

At one step the restriction is vacuous. At two steps it can be costly. The binary constructions above prove both an exact two-policy gap

$$\Gamma_2 = \frac{3}{8}\log_2 3$$

and a strict gap with the full deterministic policy family

$$\Gamma_2 \geq \frac{2}{9}.$$

Thus the response-tree requirement has genuine information-theoretic content: minimum entropy across policy marginals is not, in general, minimum entropy of a common causal realization.

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